

ECE 3436: Microprocessor Systems

Embedded ARM Cortex-M4 assembly programming on the TI TM4C123 LaunchPad, developed in Keil MDK-ARM / μ Vision.

github.com/Buggy-Scott/ECE-Archive-files — ECE 3436 MicroProcessing

119 Files in Folder	10+ Labs & Projects	8 Topic Units	TM4C123 Target MCU
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Course Overview

ECE 3436, Microprocessor Systems, is a hands-on embedded systems course focused on ARM Cortex-M4 assembly language programming and register-level hardware control. This folder of the ECE Archive collects two semesters of coursework: in-class quizzes (ICQs), homework sets, lecture slides, lab starter and completed projects, and full exam materials, all centered on programming the Texas Instruments TM4C123GH6PM microcontroller (the board used with the TI TM4C123 LaunchPad).

The work moves from number representation and basic register operations toward full hardware interaction — GPIO control and timer-driven programs — with each lab building directly on the instruction set covered in the homework immediately before it.

Skills Demonstrated

- ARM Cortex-M4 assembly programming (Keil/ARM syntax) — arithmetic, data movement, and control-flow instructions.
- Register-level microcontroller programming without relying on vendor abstraction libraries.
- Embedded toolchain use: project setup, build configuration, and flash/debug workflows in Keil μ Vision.
- Direct GPIO and SysTick peripheral configuration on a real target device (TM4C123GH6PM, Cortex-M4).
- Reading and interpreting build artifacts — map files, listing files, and linker output — to debug memory layout.

Topics Covered

Topic	What It Covers
Number Systems & Directives	Binary/hex/BCD conversions, signed representation, and Keil assembler directives (EQU, AREA, EXPORT).
Registers & Special-Purpose Registers	General-purpose register usage, the program status register, and special-purpose register access on the Cortex-M4.
Conditional Codes & Compare/Test	NZCV flag behavior, CMP/TST instructions, and conditional execution.

Topic	What It Covers
Shift, Rotate & Logical Operations	LSL/LSR/ROR and bitwise AND/OR/EOR instructions for bit-level manipulation.
Conditional Branching & Execution	Branch instructions, loop constructs, and program flow control in ARM assembly.
Data Storage: RAM & ROM	Memory-mapped data placement, LDR/STR addressing modes, and the AREA/DATA memory model.
GPIO	Direct register-level control of the TM4C123's GPIO ports for digital I/O (LED blink, switch input).
SysTick Timer	Configuring the Cortex-M4 SysTick peripheral for timed delays and periodic events.

Labs & Projects

Each lab in the archive is a complete Keil μ Vision project — source (.s), build logs, linker maps, and (where applicable) the finished .axf binary — rather than just a written report.

Lab	Focus	Description
Lab 00a–c	<i>Toolchain Setup</i>	Installing and configuring Keil MDK-ARM/ μ Vision and getting a first project building and flashing to the TM4C123 LaunchPad.
Lab 1	<i>Register-Based Statistics</i>	A from-scratch assembly program computing group means and sum-of-squared-errors entirely in registers — no memory access, pure register arithmetic and SDIV/MUL instructions. Completed with a lab partner.
Lab 2	<i>Registers & Directives</i>	Applying EQU/AREA directives and register-to-register data movement patterns from the homework unit of the same name.
Lab 3 / Directives Project	<i>Assembler Directives</i>	A structured project exercising Keil assembler directives for data and code section layout.
Lab 4	<i>Conditional Branching</i>	Implements branch-based control flow, building on the Conditional Branching & Execution class activity.
Registers 1 / 2	<i>Special-Purpose Registers</i>	Starter and completed projects working with the Cortex-M4's special-purpose registers.
RAM/ROM Starter	<i>Memory & Data Storage</i>	Starter project for the RAM/ROM data-storage unit — placing and addressing data in memory versus flash.
Shift/Rotate/Logic Activity	<i>Bit Manipulation</i>	In-class activity applying LSL/LSR/ROR and logical instructions.
GPIO Blink Starter	<i>GPIO</i>	Register-level GPIO configuration to drive an onboard LED — the entry point for all later hardware I/O labs.
Exam 1 Project	<i>Cumulative Assessment</i>	A timed programming exam implemented as a full Keil project, covering numbering, registers, and conditional logic.

Featured Lab: Register-Based Statistics Calculator

Lab 1 is a pure-register assembly program (no memory reads or writes) that computes descriptive statistics — per-group means, a grand mean, and the sum of squared errors for each group — entirely using ARM Cortex-M4 arithmetic instructions. Values are loaded as immediates, summed with ADD, divided with the Cortex-M4's hardware SDIV instruction, and squared with MUL, with results tracked across ten dedicated registers (R0–R12) without ever touching RAM.

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; Task 1 – Group Mean (register-only arithmetic)
MOV  R0, #AStudent1      ; load each group member's value
MOV  R1, #AStudent2
...
ADD  R5, R0, R1          ; running sum
ADD  R5, R5, R2
ADD  R5, R5, R3
ADD  R5, R5, R4
MOV  R6, #GroupN
SDIV R12, R5, R6         ; R12 = mean of Group A

; Task 2 – Sum of Squared Errors per group
MOV  R0, AStudent1
SUB  R1, R0, R12         ; x - mean
MUL  R1, R1, R1          ; (x - mean)^2
ADD  R8, R8, R1          ; accumulate SSE for Group A
```

This lab was completed with a partner and reflects the register-allocation discipline the course emphasizes: every intermediate result is assigned to a specific, documented register rather than spilled to memory, which is a deliberate constraint of the assignment.

Tools & Environment

- Keil MDK-ARM / μ Vision — IDE, assembler, and debugger used for every lab and exam project.
- Target device: TI TM4C123GH6PM (ARM Cortex-M4F core, 256 KB flash, 32 KB SRAM).
- CMSIS device support (TM4C123.h) for register and peripheral definitions.
- ULINK2/CMSIS-DAP flash/debug driver configuration for on-target programming.

Browse the full folder — homework, lecture slides, ICQs, and every lab project — in the repository:

<https://github.com/Buggy-Scott/ECE-Archive-files/tree/main/ECE%203436%20MicroProcessing>